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|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9 | Answer: B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|   | <p>Using <math>T^2 \propto r^3</math><br/>           (Can use in MCQ direct, but must prove starting from <math>F_g</math> provides <math>F_c</math> if in P2 or P3)</p> <p>Note to find “radius of orbit” given the satellite’s height above Earth.</p> <p>Height <math>r \Rightarrow</math> radius of orbit = <math>2r</math><br/>           Height <math>3r \Rightarrow</math> radius of orbit = <math>4r</math></p> $\left(\frac{T'}{T}\right)^2 = \left(\frac{4r}{2r}\right)^3$ $T'^2 = 2^3 T^2$ $T' = 2\sqrt{2}T$ |

|    |           |
|----|-----------|
| 10 | Answer: C |
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